

PROCESS BOOK

What makes nations *happy*?

A ranking lists them. Five years of data explain them.

782

COUNTRY-YEAR
RECORDS

5

YEARS OF DATA

8

INTERACTIVE
CHARTS

150+

COUNTRIES
TRACKED

This document describes the design process, technical decisions, and iterative development that led to the final interactive data visualisation submitted for Milestone 3. It covers preprocessing, design evolution, implementation, and individual contributions.

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1 Introduction & Vision

From a ranking to a story.

The question. What makes entire nations happier than others? The World Happiness Report surveys 150+ countries on a 0–10 life satisfaction scale and decomposes results into six measurable factors. Our goal: transform five years of this data (2015–2019) into an **accessible, story-driven visual exploration** for a general, non-expert audience, closer to data journalism than a research dashboard.

Most existing treatments present either a static snapshot or a single-variable explorer. None combine the three dimensions of *where*, *what*, and *when* into a unified narrative.

Narrative structure. The website guides users through four sections in sequence, each building on the previous. Every section pairs a primary chart with a companion that tells a genuinely different story.

Target audience. The primary audience is the general public: curious, non-expert readers who want to engage with data about human wellbeing intuitively. No statistical background is assumed. Every chart is preceded by plain-language explanatory text, and interactive features are designed to reward curiosity without requiring expertise.

Design philosophy. Three principles guided every decision throughout the project.

- **Narrative over exploration:** the user follows a curated path through the data.
- **Consistency over variety:** one color per region, one tooltip style, one transition easing.
- **Complement over duplicate:** every companion chart answers a different question from the primary chart.

Section	Question	Primary chart	Companion chart
Where	Where is happiness concentrated?	Choropleth world map	Regional strip plot
What	What factors drive it?	Stacked bar chart	Radar fingerprint
Explore	How do factors relate to scores?	Gapminder bubble chart	Factor scatter plot
When	When did things change?	Multi-line trends chart	Bar chart race

Narrative over exploration

A curated path through the data. Each section opens with a guiding question before the chart appears.

Consistency over variety

One color per world region and one shared interaction system across all charts.

Complement over duplicate

Every companion chart was chosen to reveal insights the primary chart cannot show alone.

2 Dataset & Preprocessing

Five years, 782 records, six factors.

Source. The dataset is the World Happiness Report, publicly available on [Kaggle](#) (SDSN). It consists of five CSV files, one per year from 2015 to 2019, covering 155–158 countries each year. Each file contains a happiness score and rank per country alongside six contributing factors. Scores derive from the Gallup World Poll Cantril ladder question, where respondents rate their lives from 0 (worst possible) to 10 (best possible). The five files were unified into a single `happiness.json` of **782 country-year records**.

782 unified country-year records across 5 years of World Happiness data.

Preprocessing challenges

- **Column name drift:** Family → Social support; Score → Happiness Score across years. All columns standardised to 2015 naming convention.
- **Missing Region column:** Present only in 2015–2016; re-mapped to all years via a country-level join on the 2016 file, with a manual lookup table for countries first appearing post-2016.
- **One missing value:** 2018 Trust (Government Corruption) for one country-year; left as NaN and replaced with 0 in the visualisation layer, as it contributes negligibly to that country's score.

Data pipeline

5 CSV files (2015–2019)

Raw WHR data from Kaggle

↓

Column standardisation

Names, types, region re-mapping

↓

Merge & validation

782 records, 1 NaN handled

↓

`happiness.json`

185 KB, loaded via `d3.json()`

Key EDA findings that shaped design

Finding	Design consequence
GDP ($r=0.794$), Health ($r=0.780$), Social Support ($r=0.777$) strongest. Generosity $r=0.076$ — nearly uncorrelated. 141 countries in all 5 years; 29 in fewer. Venezuela -2.10 pts; Benin $+1.54$ pts. Global average stable: $5.376 \rightarrow 5.407$.	GDP as bubble X-axis; Health as scatter default. Highlighted as “surprising” in radar & annotations. Trends & race bar filtered to full-coverage only. Anchoring facts in scrollytelling annotations. Central tension in Part 4: stability vs. turbulence.

Regional happiness averages (2019)

Region	Score
Australia & New Zealand	7.27
North America	7.09
Western Europe	6.90
Latin America & Caribbean	5.94
Eastern Asia	5.69
Central & Eastern Europe	5.57
Southeastern Asia	5.27
Middle East & North Africa	5.24
Southern Asia	4.53
Sub-Saharan Africa	4.32

What the data can and cannot claim

Happiness scores measure self-reported life satisfaction; they are not an objective welfare index. All visualisations present observed associations and comparisons, never causal recipes or predictions.

3 From Sketches to Result

From wireframe to interaction.

Part 1: Choropleth Map + Strip Plot

The **Milestone 1 sketch** was a standard choropleth with a year slider and a basic tooltip. By Milestone 3, three things had changed. The **color scale** switched from sequential blue to RdYlGn diverging: unhappy countries read red, happy ones green, which is immediately intuitive for non-expert readers.

A **country click** now opens a side panel with factor bars and a score sparkline, giving per-country stories without leaving the map. Finally, a **strip plot companion** was added: one dot per country in regional swim-lanes with a hoverable mean line, answering the question the map cannot: how unequal is happiness *within* each region?

Part 2: Stacked Bar + Radar Chart

The original sketch was a grouped bar chart, replaced in Milestone 2 by a stacked bar; because the six factors literally sum to the total score, stacking is semantically correct.

Two label bugs required fixing:

- The **SVG-embedded legend** overlapped rotated axis labels and was moved to an HTML element below the chart.
- **Label rotation** at -35° clipped long region names and was fixed with -45° , larger margins, and dx/dy offsets.

The companion chart evolved from a heatmap (same question as the bar) to a **radar fingerprint**. By normalising each axis to its global maximum, the polygon reveals a region's *shape* of happiness, not just its totals.

Evolution from Milestone 1 to Milestone 3

Where

Milestone 1: plain choropleth

Milestone 3: interactive map, side panel, strip plot companion

What

Milestone 1: grouped bar chart

Milestone 3: stacked bars, fixed labels, radar fingerprints

3 From Sketch to Result (Continued)

Interaction, animation, and storytelling.

Part 3: Bubble Chart + Factor Scatter

This section was added in Milestone 3 as the complex interactive centrepiece, inspired by Gapminder (cited in our related work). The **bubble chart** encodes GDP on X, happiness on Y, and inverse rank as size, triple encoding that lets users read wealth, standing, and happiness simultaneously.

A Play button animates 2015–2019 with smooth D3 transitions. The **factor scatter companion** uses a completely different visual grammar: regional swim-lanes on the Y-axis with a switchable factor on X (five pill buttons: Health, Social Support, Freedom, Trust, Generosity).

A vertical mean line per region enables within-vs-across comparison, a question impossible to read from the bubble chart.

Part 4: Trends Chart + Bar Chart Race

The multi-line trend chart was in the Milestone 1 sketch but evolved significantly. The original showed only the top/bottom 5 movers; the final draws all 141 full-coverage countries simultaneously, with pre-highlighted movers and a **country search box** that pins any country in blue while dimming others.

The initial companion was a slope chart (2015 vs. 2019), but it duplicated what the highlighted trend lines already showed.

It was replaced by a **bar chart race**: watching the top 15 countries jostle year by year makes rank dynamics visceral in a way no static chart can.

The Nordic bloc holds rigid; everyone else shifts around it.

Evolution from Milestone 1 to Milestone 3

Explore

Milestone 1: not planned

Milestone 3: animated bubble chart + factor scatter

When

Milestone 1: top/bottom 5 lines

Milestone 3: all 141 countries + bar chart race

4 Design & Technical Implementation

Decisions that shaped the experience.

Color system. The single most important design decision was assigning one fixed color to each of the 10 world regions and using it consistently across all eight charts. Once a user learns that Western Europe is blue and Sub-Saharan Africa is orange-red, that knowledge transfers without re-learning.

• Western Europe	#2563EB
• North America	#7C3AED
• Australia & New Zealand	#0891B2
• Central & Eastern Europe	#65A30D
• Eastern Asia	#D97706
• Southeastern Asia	#F59E0B
• Latin America & Caribbean	#DC2626
• Middle East & N. Africa	#9333EA
• Southern Asia	#0D9488
• Sub-Saharan Africa	#EA580C

Warm background & typography.

Parchment background (#FDF8EE), near-black text (#1A1207), warm muted labels (#4A3F2E). This palette gives the site a humanistic, journalistic feel consistent with its editorial tone, closer to a magazine than a dashboard.

Shared interaction grammar. All eight charts share identical patterns. Tooltips use a single `WHR.createTooltip()` factory: dark background, bold country name, light-blue secondary info. Mean lines carry a transparent 16px hit area on top of the visible 2.5px stroke. Transitions use `easeQuadInOut` at 500–750ms. The three Play buttons share the same step-timer pattern internally.

Scrollytelling annotations. Each section reveals one curated insight via `IntersectionObserver` on first scroll: e.g. “Finland, Denmark & Norway have held the top 3 spots every year.” One sentence, emoji anchor, appears once per session.

Architecture. Fully client-side: no build step, no framework, no server beyond a static host. Vanilla D3 v7. Three CDN dependencies: D3 v7, TopoJSON v3, World Atlas 110m. Ten JS files coordinated by a shared WHR namespace in `utils.js`. Storing data as a single `happiness.json` (185 KB) avoids any server-side parser and makes `d3.json()` the only data dependency.

The WHR namespace. The key refactoring decision: extracting shared code into a WHR global. Before this, four files each defined their own tooltip creation, mousemove handler, and grid line setup. The namespace provides `FACTORS`, `REGION_COLORS`, `createTooltip()`, `moveTooltip()`, `hideTooltip()`, `buildSVG()`, `addGridLines()`, `fullCoverageCountries()`, and `computeChanges()`. A single fix in `utils.js` now propagates to all charts.

Key technical challenges.

Off-screen width bug: `getBoundingClientRect()` returns 0 for unpainted elements. Charts below the fold rendered with zero SVG width. Fixed by wrapping all chart initialisations in `requestAnimationFrame()` in `main.js`, with an `offsetWidth` fallback in `buildSVG()`.

Country name mismatch: TopoJSON spellings differ from the CSV (e.g. “United States of America” vs. “United States”). A `Country_map` field bridges the gap; `findRecord()` checks both fields on every hover and click.

Missing Region (2017–2019): The `Region` column is absent from 3 of the 5 source files. Solved by building a lookup table from the 2016 file and merging it into the unified JSON during preprocessing.

Label clipping: Long region names clipped at the SVG left edge at -35° rotation. Fixed with -45° , larger bottom margin, and `dx/dy` offsets on the bar chart; larger radius and outer label padding on the radar.

5 Challenges & Iterations

Every pivot made the story cleaner.

“What can this chart show that the primary chart cannot?”

This became the governing principle behind every chart pairing and every redesign decision in the project.

The key lesson

A companion chart that asks the same question as the primary chart adds visual complexity without adding insight. Every replacement below came from identifying redundancy and replacing it with a chart that opened a genuinely new question.

The six pivots fell into two categories:

- **Chart replacements:** heatmap, slope chart, ranked bar chart.
- **Layout/rendering fixes:** radar layout, SVG legend, label rotation.

The heatmap, slope chart, and ranked bar chart were all implemented before being discarded. Building and discarding was not wasted effort: each rejected chart clarified what the final chart needed to achieve.

Testing & debugging

The label clipping and zero-width bugs only appeared during browser testing at realistic viewport sizes, not during development in a maximised window.

Both fixes required understanding browser rendering timing and SVG coordinate systems rather than D3 itself.

The most difficult bug was the zero-width rendering issue: `getBoundingClientRect()` returned 0 for charts below the fold before paint. The final solution was wrapping chart initialisation in `requestAnimationFrame()`.

Major pivots

Initial idea	Final decision
Heatmap (regions × factors)	Radar chart showing regional “fingerprints”.
Slope chart (2015 vs. 2019)	Bar chart race showing rank dynamics over time.
Ranked top-20 bar chart	Strip plot revealing within-region distributions.
Radar in 2-column layout	Full-width radar below stacked bar chart.
SVG-embedded legend	External HTML legend below chart.
−35° label rotation	−45° rotation with offsets and larger margins.

Final JS structure

File	Responsibility
<code>utils.js</code>	Shared namespace, helpers, SVG scaffolding
<code>map.js</code>	Map, detail panel, sparkline
<code>factors.js</code>	Stacked factor bars
<code>radar.js</code>	Regional radar chart
<code>bubble.js</code>	Gapminder-style bubble chart
<code>trends.js</code>	Multi-line trend chart
<code>raceBar.js</code>	Animated bar chart race
<code>main.js</code>	Data loading and chart initialisation

6 Peer Assessment & Conclusion

What we built, and who built what.

Contribution breakdown

Member	Contributions
Dafer Yassine	Lead on data preprocessing and EDA (Milestone 1). Built the choropleth map, detail panel, and sparkline. Coordinated the GitHub repository and README. Wrote the process book.
Ali El Bouiri	Built the stacked factor bar chart and radar fingerprint chart. Responsible for the CSS design system (color variables, warm palette, typography). Handled label clipping bugs and layout fixes.
Mohamed Chafai El Alaoui	Built the Gapminder bubble chart and factor scatter dot plot. Implemented the Play/Pause animation system shared across three charts. Wrote <code>utils.js</code> and led the code cleanup refactor.
Dafer Mohamed Amine	Built the multi-line trends chart with search and pin. Built the strip plot and bar chart race. Implemented scrollytelling annotations via <code>IntersectionObserver</code> . Recorded and edited the screencast.

Conclusion

The World Happiness Report dataset is widely explored, but almost always as a static ranking or a single-variable scatter plot. Our goal was different: to turn five years of data into a **narrative journey** that a non-expert could follow from geography to factors to dynamics, with every chart earning its place by answering a question the previous one could not.

The final website delivers eight interactive charts across four sections, all sharing a consistent visual language: one color per region, one tooltip style, one interaction grammar. The WHR shared namespace ensures that a fix or improvement propagates to every chart simultaneously. The warm parchment palette and plain-language section introductions keep the tone humanistic rather than clinical.

The process was iterative. Six charts were built and replaced; two label bugs required understanding browser rendering internals; the companion chart for every section went through at least one rejected alternative. What remained after each iteration was simpler, more focused, and more honest about what the data could actually show.

The most important lesson of the project is also its most transferable: **a good visualisation does not show everything the data contains, it shows one thing clearly, and makes the reader want to ask the next question.** Each section of the website is designed to end with exactly that impulse.